

Supplementary Information for Knowing When Not to Publish Scientific AI Candidates

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Supplement A: Formal definitions and proofs

For a frozen generated universe $\mathcal{P}_{\text{full}} = \{p_1, \dots, p_N\}$, each candidate unit has a score S_p , a latent validity label $Y_p \in \{0, 1\}$, and an observed one-sided audit or official-support label $A_p \in \{0, 1\}$. Each certification row fixes a candidate budget K and evaluates the corresponding top- K run universe. Below, $\mathcal{P}$ denotes that evaluated run universe unless stated otherwise, so $|\mathcal{P}| \leq K$. The one-sided condition is $A_p = 1 \Rightarrow Y_p = 1$. For calibration video i , define the true false set and the null superset

$$\mathcal{F}_i = \{p : Y_p = 0\}, \quad \mathcal{N}_i = \{p : A_p = 0\}.$$

By one-sided reliability, $\mathcal{F}_i \subseteq \mathcal{N}_i$. Let

$$M_i^F = \max_{p \in \mathcal{F}_i} S_p, \quad B_i^{\text{ns}} = \max_{p \in \mathcal{N}_i} S_p,$$

with the convention that empty maxima follow the policy specified in the main text. The main text distinguishes the full generated universe size N from the predeclared candidate budget K . In this supplement, $\mathcal{P}$ denotes the evaluated run universe after the fixed budget/ranking rule has been applied, and we write $M = |\mathcal{P}| = K$ for the number of hypotheses entering the self-consistency denominator. When implementation logs use m , it is the same run-level quantity.

The candidate universe is conditional throughout. All validity statements are made after fixing the proposal backend, prompt/query set, linking rule, score rule, release grid, calibration/test split and candidate budget. This conditioning is essential: PARC certifies a release decision for a specified finite proposal universe, not discovery over all possible unknown objects.

Lemma 1 (Null-superset dominance). *For every calibration block with non-empty false and null-superset blocks,*

$$B_i^{\text{ns}} \geq M_i^F.$$

Proof. The result follows directly from $\mathcal{F}_i \subseteq \mathcal{N}_i$: the maximum over a superset cannot be smaller than the maximum over a subset. $\square$

Theorem 1 (Block p-value validity). *Assume (i) calibration blocks and the test block are exchangeable under the domain block construction, (ii) one-sided audit reliability holds, $A_p = 1 \Rightarrow Y_p = 1$, and (iii) the candidate generator and score are frozen before calibration and testing. For every actual false test candidate p , the null-superset block p-value*

$$q_p = \frac{1 + \sum_{i=1}^n \mathbf{1}[B_i^{\text{ns}} \geq S_p]}{n + 1}$$

is super-uniform: $\mathbb{P}(q_p \leq u) \leq u$ for all $u \in [0, 1]$.

Table 1: Notation used throughout the Supplementary Information.

Symbol	Meaning
$\mathcal{P}$	Frozen run-level candidate universe for the evaluated split after the fixed proposal, linking, ranking and candidate-budget rules are applied. In the main text this budget is denoted K ; here $M = \mathcal{P} = K$.
p	Candidate unit: a cell link, building link, detection, object path or tracklet, depending on the domain. A unit has block id, compatibility fields and frozen score.
S_p	Frozen unit score used for release calibration. It is fixed before calibration/test evaluation and never uses test labels.
Y_p	Latent unit validity: $Y_p = 1$ denotes a valid candidate and $Y_p = 0$ denotes an actual false release candidate.
A_p	One-sided observed support/audit label. The only structural assumption is $A_p = 1 \Rightarrow Y_p = 1$.
$\mathcal{F}_i$	Unobserved false units in calibration block i .
$\mathcal{N}_i$	Null-superset units in calibration block i , obtained by removing only verified positives and retaining all unverified candidates.
B_i^{ns}	Null-superset block maximum for calibration block i .
q_p, q_p^{any}	Block p-value and release-grid p-value for candidate unit p .
E_p	E-value obtained by applying an e-calibrator to q_p^{any} .
$\mathcal{R}$	Data-dependent released set returned by SCS-Greedy.
α	Target false-release fraction risk.

Proof. First consider the oracle false-only p-value formed by replacing B_i^{ns} with M_i^F and including the test false block in the exchangeable sequence. The rank argument for exchangeable block maxima gives a super-uniform p-value for a false test candidate. By the dominance lemma, replacing each M_i^F by B_i^{ns} can only increase the number of calibration maxima exceeding S_p , and therefore cannot decrease the p-value. Thus the null-superset p-value is at least as conservative as the false-only p-value. $\square$

For a release grid $\mathcal{L}$ with weights $w_j > 0$ and $\sum_j w_j \leq 1$, define

$$q_p^{\text{any}} = \min_{j: L_j \leq L_p} \left(\frac{q_{p,j}}{w_j} \right) \wedge 1.$$

The weighted-min construction is valid by the union bound:

$$\Pr(q_p^{\text{any}} \leq u) \leq \sum_j \Pr(q_{p,j} \leq u w_j) \leq u \sum_j w_j \leq u.$$

If $f_\gamma(u) = \gamma u^{\gamma-1}$ with $0 < \gamma < 1$, then $E_p = f_\gamma(q_p^{\text{any}})$ is an e-value for every false path because a super-uniform p-value U satisfies

$$\mathbb{E}[f_\gamma(U)] \leq \int_0^1 \gamma u^{\gamma-1} du = 1.$$

Theorem 2 (Self-consistent FTR control). *Let $\mathcal{H}_0 \subseteq \mathcal{P}$ be the false candidates. Assume every false candidate has an e-value E_p with $\mathbb{E}[E_p] \leq 1$. If the selected set R satisfies*

$$p \in R \Rightarrow E_p \geq \frac{K}{\alpha(|R| \vee 1)},$$

Table 2: Core assumptions and where they enter the proof.

Assumption	Role	Main consequence
Frozen proposal universe	Candidate generation, scoring, release grid and budget are fixed before calibration/test evaluation.	Conditions the hypothesis set used in SCS.
Block-level exchangeability	Calibration blocks and the test block are exchangeable under the domain block construction.	Enables the rank p-value proof.
One-sided positive reliability	$A_p = 1 \Rightarrow Y_p = 1$.	Ensures false candidates are contained in the null superset.
Coverage-conditional exchangeability	Covered calibration blocks and covered test false blocks remain exchangeable after conditioning on coverage.	Enables the paperfacing covered-regime policy.
Graph compatibility is frozen	Detection-node conflicts and protected components are defined before selection.	Allows data-dependent greedy selection under final self-consistency.

then

$$\mathbb{E} \left[\frac{|R \cap \mathcal{H}_0|}{|R| \vee 1} \right] \leq \alpha.$$

Proof. For any false path p , self-consistency implies

$$\frac{\mathbf{1}[p \in R]}{|R| \vee 1} \leq \frac{\alpha}{K} E_p.$$

Summing over $p \in \mathcal{H}_0$ and taking expectations gives

$$\mathbb{E}[\text{FDP}(R)] \leq \frac{\alpha}{K} \sum_{p \in \mathcal{H}_0} \mathbb{E}[E_p] \leq \frac{\alpha}{K} |\mathcal{H}_0| \leq \alpha,$$

because $|\mathcal{H}_0| \leq K$. The proof allows R to be data-dependent; it only uses the final self-consistency inequality. $\square$

If no non-empty compatible set satisfies the self-consistency inequality, the empty release is returned. Its false-release proportion is zero by definition, so refusal is a valid certified decision.

Empty-block policies. The conservative empty-block policy sets $B_i^{\text{ns}} = +\infty$ whenever the calibration block has no covered null-superset candidate. This yields a marginally valid but often powerless p-value. The coverage-conditional policy instead conditions on the event that the cell/checkpoint is covered and applies the rank argument within the covered regime. Its additional assumption is conditional exchangeability of the covered calibration block maxima and the covered test false block. When this assumption is not defensible, the conservative empty-block policy remains the fallback.

Finite resolution. With n covered calibration videos and G release checkpoints, the smallest Bonferroni-weighted release-grid p-value is on the order of $G/(n+1)$. This caps the maximum attainable e-value for a fixed calibrator. Even when the cap is high enough, release still requires enough high-evidence candidates to satisfy self-consistency. The mass statistic $\max_k \alpha k E_{(k)}/K$ characterizes the unconstrained self-consistency frontier before graph conflicts.

Table 3: Validity-power tradeoff for empty-block policies.

Policy	Validity statement	Practical role
Conservative $+\infty$ block	Marginally valid under video-level exchangeability and one-sided positives.	Always available, often powerless when many blocks are empty.
Coverage-conditional block	Valid in the covered regime under conditional exchangeability.	Paper-facing setting for non-empty release.
Diagnostic best- K grid	No deployment guarantee unless selected by a tuning-fixed rule.	Used only to understand power/frontier behaviour.

Why finite resolution can force refusal. Let $q_{\min}$ be the smallest attainable release-grid p-value after weighting. For a power e-calibrator $f_{\gamma}(u) = \gamma u^{\gamma-1}$, the largest attainable e-value is

$$E_{\max} = f_{\gamma}(q_{\min}).$$

If $E_{\max} < K/\alpha$, then even a single path cannot cross the $k = 1$ self-consistency threshold. If $E_{\max}$ is above K/α , a non-empty release is still not automatic: the selected set of size k requires at least k mutually compatible candidates above $K/(\alpha k)$. The feasibility mass

$$\Phi = \max_{k \geq 1} \frac{\alpha k E_{(k)}}{K}$$

is therefore a pre-selection characterization of the uniform self-consistency frontier. When $\Phi < 1$, no non-empty uniform self-consistent release exists before considering graph conflicts. When $\Phi \geq 1$, the graph selector may still return a smaller set, or return empty if all high-evidence candidates conflict, but the evidence mass itself is sufficient.

Coverage-conditional scope. The paper-facing covered-regime rows use the coverage-conditional policy. The guarantee should be read literally: it is a guarantee inside the covered regime induced by the fixed generator, cell/checkpoint definition and candidate budget. It is not a marginal guarantee over videos or cells that contain no relevant candidate coverage. The conservative $+\infty$ policy remains the marginal fallback, and its lower power is reported to make the validity-power tradeoff explicit.

Supplement B: Algorithms

The implementation is a four-stage post-hoc certification pipeline. First, a frozen proposal source exports a finite candidate universe. Second, calibration videos define null-superset block maxima after removing only verified positives. Third, release-grid p-values are converted into e-values. Fourth, SCS-Greedy selects a compatible release set or returns a valid empty release.

Table 4: Pipeline stages and label access.

Stage	Operation	Label access
Proposal export	Generate candidate units, scores, blocks, optional nodes/edges and conflicts.	No audit or test labels.
Calibration	Remove verified positives and compute null-superset block maxima.	Calibration one-sided labels only.
E-value construction	Convert release-grid p-values into candidate e-values.	No test labels.
SCS release	Greedily select compatible candidates satisfying self-consistency.	No labels.
Diagnostics	Report UTR, audited FTR and conservative FTR.	Evaluation/audit labels after release.

Algorithm B1: PARC certification pipeline.

Input:

```
frozen candidate universes for tuning, calibration and test videos
one-sided support/audit labels on calibration videos
fixed candidate budget M, release grid L, risk level alpha
e-calibrator f_gamma and compatibility predicate compatible(.,.)
```

Output:

```
released set R_out or empty certified refusal
```

1. Freeze the proposal backend, linking/conflict rule, score rule and grid.
2. For every calibration block i and grid checkpoint L_j :
 remove only verified positives;
 retain all unverified candidates as the null superset $N_{\{i,j\}}$;
 store $B^{\{ns\}}_{\{i,j\}} = \max_{\{p \in N_{\{i,j\}}\}} S_{\{p,j\}}$.
3. For every test candidate unit p :
 compute block p-values $q_{\{p,j\}}$;
 combine grid checkpoints into $q_p^{\{any\}}$;
 convert $q_p^{\{any\}}$ to an e-value E_p .
4. Run SCS-Greedy on test candidates, E-values and the compatibility graph.
5. Report R_{out} , or the empty release if SCS-Greedy returns no non-empty set.
6. Compute UTR, audited FTR and conservative FTR only after release.

Algorithm B2: Null-superset block construction.

```
Input: calibration block i, checkpoint L_j, frozen score S_p, labels A_p
N <- empty set
for candidate unit p in calibration block i:
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```

    if unit p is eligible at checkpoint L_j:
        if A_p != 1:
            add p to N
if N is non-empty:
    return max_{p in N} S_p
else if conservative empty-block policy:
    return +infinity
else if coverage-conditional policy:
    mark cell/checkpoint as uncovered and exclude from covered-rank set

```

Algorithm B3: Release-grid e-values.

```

Input: test candidate p, calibration block maxima  $B^{\text{ns}}_{\{i,j\}}$ ,
      checkpoint weights  $w_j$ , calibrator  $f_\gamma$ 
for every eligible checkpoint  $L_j \leq L_p$ :
     $q_{\{p,j\}} \leftarrow (1 + \text{number of calibration blocks with } B^{\text{ns}}_{\{i,j\}} \geq S_{\{p,j\}}) / (n_j + 1)$ 
 $q_{p\_any} \leftarrow \min_j q_{\{p,j\}} / w_j$ , clipped at 1
 $E_p \leftarrow f_\gamma(q_{p\_any})$ 
return  $E_p$ 

```

Main release grid and e-calibrator choices. The main protocol uses a single 2.0 s release checkpoint:

$$\mathcal{L} = \{2.0s\}, \quad w_1 = 1.$$

This means the main table does not rely on a long anytime grid; the release-grid notation is kept because the implementation and proofs support multiple checkpoints. The power e-calibrator is $f_\gamma(u) = \gamma u^{\gamma-1}$. For the frozen experiment rows, γ is selected before test evaluation from finite-resolution information using the tuning rule

$$\gamma^* = -\frac{1}{\log r},$$

where r is the smallest attainable weighted rank p-value for the covered calibration block. No test audit labels, released-set labels or conservative diagnostics are used to choose γ . The selected γ values are row-specific because covered calibration denominators differ by dataset, source and seed; the CTC and SpaceNet 7 implementation rows use $\gamma = 0.229$ and $\gamma = 0.235$, respectively, and the materials row records its analogous finite-resolution choice in the provenance sidecars. The choice is the finite-resolution optimum within the power-calibrator family: for smallest attainable p-value r , the derivative of $\log f_\gamma(r) = \log \gamma + (\gamma - 1) \log r$ vanishes at $\gamma^* = -1/\log r$, yielding $f_{\gamma^*}(r) = 1/(er[-\log r])$. Gamma sensitivity beyond this finite-resolution rule is a power diagnostic, not a validity requirement: every fixed $0 < \gamma < 1$ remains an e-calibrator.

SCS-Greedy is the selector used in the main experiments. It enforces graph compatibility but does not need to solve an optimal path-packing problem for the risk guarantee.

```

Input: candidates P, e-values E_p, utilities U_p,
      compatibility predicate compatible(.,.), risk level alpha
M <- |P|
for k = M, M-1, ..., 1:

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tau <- M / (alpha * k)
C <- {p in P : E_p >= tau}
sort C by decreasing utility U_p
R <- empty set
for p in C:
  if p is compatible with every candidate in R:
    R <- R union {p}
  if |R| = k:
    assert every p in R satisfies E_p >= M / (alpha * |R|)
    return R
return empty set

```

The implementation uses domain-specific conflicts supplied by the frozen candidate graph. For links these conflicts are mutually inconsistent identity assignments; for detections they are duplicate detections when exported; for tracks or paths they include shared detection nodes, duplicate hypotheses and protected components when present in the graph. Candidate utilities are frozen before calibration and never use audit or official-support labels. For m scanned candidates and q conflicts, the naive scan is $O(m^2 + q)$ per risk level; the released experiments are much smaller than the regime where a specialized maximum independent-set approximation is needed.

Table 5: Compatibility predicate used by SCS-Greedy.

Constraint	Interpretation	Frozen?
Identity/link conflict	Two links cannot both be released if they assign the same object to inconsistent parents or successors.	yes
Detection-node conflict	Two paths cannot both be released if they share the same underlying detection node at a frame.	yes
Duplicate hypothesis	Near-duplicate candidates from the same proposal component are treated as mutually incompatible when exported as conflicts.	yes
Protected component	Optional connected components can be protected to avoid releasing mutually inconsistent alternatives.	yes
Temporal eligibility	A candidate must be eligible for the release checkpoint used to construct its e-value.	yes
Label information	Audit/support labels are not used by SCS-Greedy; they enter calibration construction and post-release diagnostics only.	fixed

Refusal condition. SCS-Greedy returns an empty release when no scanned release size k yields k mutually compatible candidates satisfying $E_p \geq K/(\alpha k)$. This is a certified refusal, not a declaration that all candidates are false. The empty release is risk-valid by construction, and the accompanying diagnostics identify whether refusal is due to low evidence mass, sparse coverage or graph conflicts.

Greedy optimality scope. The formal certificate depends only on the final returned set satisfying self-consistency and compatibility. It does not require SCS-Greedy to find the largest compatible self-consistent set. An exact maximum-independent-set or ILP feasibility oracle is a power diagnostic: it can show whether the deployed polynomial selector missed a larger releasable set, but it does not change the validity proof for the set actually returned. The selector-optimality diagnostics included here apply this oracle only to scoped refusal rows where the public package

provides the required aggregate or compatibility information. For those rows, the reported refusals are finite-resolution or pre-graph evidence-mass failures and the ILP oracle is not feasible; we therefore do not attribute those empty outputs to SCS-Greedy compatibility loss. The paper still does not claim a general selector approximation ratio or global optimality theorem. This limitation is separated from refusal validity: if SCS-Greedy returns a non-empty self-consistent set, the set is certified; if it returns empty outside the oracle-diagnosed rows, the refusal should be read as a deployed-selector and evidence-mass diagnostic rather than as a mathematical impossibility result for every conceivable graph solver.

Supplement C: Main protocol, data provenance and baselines

This note documents the exact scope of the main protocol and explains why some rows are reported as main results while others are retained as appendix or provenance-only evidence. The central manuscript claims are made on non-oracle partial-verification rows for CTC biomedical cell links, retrospective WBM materials-screening candidates, a project-audited iWildCam animal-present ecology row, and a SpaceNet 7 same-building release/refusal check. Rows with $\rho = 1.0$ are full-information oracle diagnostics, not primary evidence. The iWildCam species-level and strict $\alpha = 0.10$ animal-present rows remain operating-envelope diagnostics, and OVT-B/TAO/BURST/LVIS rows are scoped open-world perception generality evidence.

Table 6: Scientific-AI main protocol scope. Detailed per-seed rows are supplied as Supplementary Data.

Item	Main setting	Notes
Primary domains	CTC, WBM materials screening, iWildCam and SpaceNet 7	Link-level release in CTC/SpaceNet, crystal-candidate release in WBM and animal-present detection release in iWildCam.
Verification fraction	$\rho < 1$	$\rho = 1.0$ is oracle diagnostic only.
Risk levels	strict $\alpha = 0.10$, operational $\alpha = 0.20$	Learned-hybrid CTC supplies strict controlled release; CGCNN supplies the predeclared materials calibration endpoint; ALIGNN-FF supplies requested-budget stopping and fixed-budget utility evidence. iWildCam supplies an operational project-team-audited ecology release; SpaceNet remains a relaxed-risk release/refusal check.
Seeds	0–19	Reported as non-empty seeds and mean release size.
Primary row rule	predeclared by ρ , K , α and non-empty seeds	Held-out FTR is reported after row selection, not optimized.
Boundary/general rows	iWildCam strict/species rows and open-world perception	Scope-marked as operating-envelope or generality evidence.

Main-table inclusion criteria. A CTC, materials, iWildCam or SpaceNet 7 row appears in the primary quantitative table only if it has a frozen candidate universe, a documented proposal source, seeds 0–19, a fixed candidate budget and risk level, and post-release FTR computed against the declared held-out, public-DFT or human-audit source. The strict calibration target is $\alpha = 0.10$ with at least 18/20 non-empty seeds. The learned-hybrid CTC source meets this target as the strict controlled anchor, and the public CGCNN WBM source meets it as the predeclared materials calibration endpoint; the ALIGNN-FF WBM row is reported as materials source-sensitivity utility evidence because it shows requested-budget certified stopping and fixed-budget follow-up utility relative to full raw top- K release at larger budgets. When this target is not met, predeclared $\alpha = 0.20$ rows may be reported only as relaxed-risk evidence. Rows using $\rho = 1.0$, iWildCam species-level or strict-risk boundary diagnostics, LVIS detection, mask-path scaffolds or OVT-B/TAO/BURST tracking are scope-marked outside the primary scientific-domain claim.

Table 7: Data-provenance policy.

Artifact class	Included in review package	Excluded from review package
Derived tables	Paper-facing CSV summaries, per-seed table inputs, figure inputs and provenance sidecars.	None.
Audit data	Audit summaries, label-count tables, Boundary-500/Audit2000 reannotation summaries and SHA256 manifests.	Raw private montage browsing cache if it contains benchmark frames.
Candidate outputs	Public-safe candidate summaries and release diagnostics.	Raw frames, detector caches, model weights and benchmark videos.
Code	Certification, table-generation and plotting scripts; tiny fixtures and validation commands.	Scripts that require private credentials or raw benchmark redistribution.
Benchmarks	Dataset names, split definitions and license pointers.	Raw OVT-B, TAO, BURST and LVIS files.

CTC learned-hybrid strict-risk source. The learned-hybrid CTC source is sequence-disjoint: sequence 01 is used to train the scorer and sequence 02 is used for held-out PARC certification. The frozen score combines geometry with crop appearance and correlation signals, and GT/matching leakage columns are excluded from the model report. The held-out sequence-02 universe contains 146,230 candidate links and 29,306 GT-supported links. Under $\rho = 0.10$, top-score partial verification and $\alpha = 0.10$, $K \in \{10, 25, 50, 75, 100, 300\}$ releases in 20/20 calibration-mask seeds with held-out FTR 0.000. The $K = 100$ row has maximum observed e-value 13.378, required e-value 10.0 and best mass ratio 1.338. The reverse split trains on sequence 02 and certifies on held-out sequence 01, again releasing in 20/20 calibration-mask seeds with held-out FTR 0.000 across the same K grid. The random-score negative control preserves the held-out candidate universe, labels and blocks but destroys ranking evidence; raw top- K FTR is 0.76–0.81 and PARC refuses every row at $\alpha = 0.10$ and $\alpha = 0.20$. The leakage audit records no train/evaluation sequence overlap, no GT or matching-label fields in the score, training-sequence-only normalization and a scorer frozen before PARC calibration. This row is used as the strict AI-assisted release anchor; the geometric CTC row is retained as a transparent structured-source control.

Materials-screening strict-risk source. The materials-screening row uses public Matbench Discovery/WBM candidates as a retrospective release-certification task. The release unit is one WBM unique prototype and the positive event is DFT stability, $e_{\text{above hull}} \leq 0$ eV per atom. The public CGCNN 10-member ensemble supplies a frozen learned ranking over 215,486 candidates; DFT stability labels are masked for calibration and used only after release to compute actual FTR. The primary block definition groups candidates by reduced-composition family, with exact-composition and Wyckoff-aware rows retained as sensitivity diagnostics. At the primary endpoint $\rho = 0.10$, $\alpha = 0.10$, $K = 100$, PARC releases in 20/20 calibration-mask seeds with mean release 100 and held-out FTR 0.030. The $K = 300$ row also releases in 20/20 calibration-mask seeds but has mean FTR 0.0916 and is reported as a larger-volume sensitivity, not as the primary materials claim. Random-score controls at $\alpha = 0.10$, $K = 300/1000$ are refused despite raw top- K FTR about 0.85, and the $K = 5000$ CGCNN request is refused where raw top- K FTR is 0.4065. Modern-model sensitivity uses a public ALIGNN-FF WBM prediction source on the same candidate universe and

blocks. At $\alpha = 0.10$, $K = 300$, raw top- K FTR is 0.253 while PARC FTR is 0.087; at $K = 500$, raw top- K FTR is 0.327 while PARC FTR is 0.048. This row is reported as source-sensitivity utility evidence for the materials claim, not as a replacement for the predeclared CGCNN calibration endpoint. The row is a retrospective certificate for a public WBM/DFT benchmark slice; it is not a claim of new materials discovery, prospective synthesis or experimental synthesizability. The block-size heterogeneity package adds candidate-level materials reruns in which block maxima are size-matched or downsampled before certification. These reruns retain the qualitative release/refusal pattern or expose conservative power loss; CTC and SpaceNet are explicitly limited to aggregate or audit-sample diagnostics because the public paper package does not redistribute candidate-level artifacts needed for an equivalent rerun. The fixed-budget utility package reports the same materials rows as downstream follow-up decisions: unstable follow-ups prevented, cost per true candidate and high-volume refusal value. At $\alpha = 0.10$, $K = 500$, the CGCNN, ALIGNN-FF and MEGNet certified queues prevent 158.25, 158.3 and 105.4 unstable follow-ups per seed, respectively; at $K = 5000$, refusal prevents 2,032.6, 2,577.4 and 2,683.5 unstable follow-ups per seed. The prospective-materials package is protocol-only: A1 temporal splitting requires an external label-release timestamp or snapshot, A2 independent DFT cross-validation requires an external source-join table, and the frozen A3 DFT follow-up pilot remains blocked until an unlabeled generated candidate pool passes public-label exclusion and novelty crossmatch. None of these prospective protocols is reported as completed positive evidence.

Table 8: Seed variability and FTR uncertainty for the primary rows. Intervals are seed bootstraps over the 20 fixed split seeds; they summarize run-to-run variability and are not independent Bernoulli confidence intervals over released links.

Domain/source	Role	ρ	K	α	Non-empty	Release mean [range; CI]	FTR mean [max; CI]	Raw top- K FTR
CTC learned-hybrid	strict controlled release	0.10	100	0.10	20/20	100.00 [100-100; 100-100]	0.000 [0.000; 0.000-0.000]	0.000
Materials CGCNN	calibration release row	0.10	100	0.10	20/20	100.00 [100-100; 100-100]	0.030 [0.100; 0.016-0.045]	—
CTC geometric linker	structured-source control	0.10	100	0.20	20/20	100.00 [100-100; 100-100]	0.000 [0.000; 0.000-0.000]	0.000
SpaceNet geometry linker	relaxed masked-label operating check	0.10	100	0.20	17/20	81.75 [0-100; 65.35-95.85]	0.003 [0.010; 0.001-0.005]	0.003
SpaceNet randomized linker	unsafe-source refusal	0.10	100	0.20	0/20	0.00 [0-0; 0-0]	0.000 [0.000; 0.000-0.000]	0.6575

SpaceNet 7 human-audit operating check. In addition to the masked-label SpaceNet rows, the artifact includes a protocolized real-audit loop for same-building links. The calibration audit inspected 800 high-score candidates and supplied one-sided verified positives; unconfirmed candidates were retained as unsupported rather than used as trusted negatives. The pre-registered real-audit operating point was $K = 100$, $\alpha = 0.20$, which produced certified refusal in 20/20 fixed-protocol audit seeds. The failure diagnostic reports a mean required e-value of 5.0, a mean maximum observed e-value of 6.061, and a mean best mass ratio of 0.635; the dominant failure mode was insufficient high-evidence mass for uniform SCS. A pre-specified $K = 50$, $\alpha = 0.20$ diagnostic row produced non-empty releases in 18/20 fixed-protocol audit seeds with mean release size 43.95. Project-team blind visual review of the 147 unique released candidates found 147 same-building links, no false links and no uncertain labels, giving visual-audit FTR 0.000 and uncertain-as-false FTR 0.000. Official metadata is reported only as a proxy diagnostic and is not substituted for the visual-audit FTR.

SpaceNet 7 human-audit files. The complete real-audit package includes $K = 50$ release-audit labels, $K = 100$ failure diagnostics, block coverage, raw top- K audit, calibration-audit summaries and per-seed rows. Exact paper-facing filenames are listed in the Supplementary Data index. The release-audit gate is human-confirmed; auxiliary calibration-coverage and raw-top- K files retain

Table 9: SpaceNet 7 real-audit operating checks. The $K = 100$ row is the pre-registered human-audit request and is interpreted as certified refusal. The $K = 50$ row is a diagnostic low-volume release confirmed by project-team blind visual review, not a replacement for a $K = 100$ deployment claim.

Regime	Status	K	α	Non-empty	Mean release	Audit/FTR outcome	Interpretation
Primary human-audit request	real-audit refusal	100	0.20	0/20	0.00	–	audited calibration signal did not support original volume
Diagnostic low-volume row	human-confirmed release	50	0.20	18/20	43.95	human FTR 0.000; $n = 147$; uncertain 0	lower-volume diagnostic release

their raw review-status fields and are used as protocol diagnostics rather than as independent second-review reliability claims.

iWildCam human-audit second review. The iWildCam operational ecology row now includes a formal human second review. The reviewed set contains all 167 released candidates, all 586 calibration not-animal rows, a random 300 calibration animal rows and all 300 raw top- K rows. Across 1,123 rows, the second review has 110 disagreements with the primary audit, label agreement 0.902, verified-positive agreement 0.902 and Cohen’s $\kappa = 0.804$ with bootstrap 95% interval [0.768, 0.838]. The release-audit subset remains 167/167 animal-present under the second review, while disagreements are concentrated in calibration-review candidates and are handled conservatively for verified-positive use.

Raw top- K releases the highest-scoring K candidates without a certificate. The fixed score threshold releases candidates above a pre-specified score cutoff. The per-source calibrated threshold uses the tuning split to choose a score threshold for each proposal source. The split conformal p-value threshold releases candidates whose candidate-level p-values pass a nominal threshold but does not account for set-level false-release fraction risk. The post-filter e-value rule thresholds candidates and then reports risk diagnostics without the SCS condition. The e-BH-style row applies an e-value multiple-testing rule without enforcing the domain compatibility and partial-verification structure used by PARC. Equivalently, PARC’s final self-consistency threshold can be read as an e-BH-style denominator applied to a candidate release set; the difference is that PARC first constructs valid e-values from a one-sided null superset and then applies the compatibility-aware release/refusal rule. The oracle true upper bound uses audit/latent labels for a diagnostic upper bound and is not a deployable method.

Exact baseline rules. Raw top- K sorts the frozen candidate universe by the same frozen utility used by the selector and releases the first K candidates, ignoring graph compatibility and risk calibration. Fixed thresholding uses a pre-specified detector-score cutoff; it is deployable but has no finite-sample false-release certificate under partial verification. The per-source calibrated threshold chooses a score cutoff separately for each proposal source on tuning/calibration material, then releases above-threshold candidates in the test universe. The split-conformal threshold converts candidate scores into p-values but thresholds individual candidates rather than controlling the set-level false-release rate. The post-filter e-value baseline constructs e-values but releases by an individual evidence cutoff without the SCS denominator $K/(\alpha|\mathcal{R}|)$. The e-BH-style selection row follows an e-value multiple-testing flavour but does not enforce the domain compatibility and final self-consistency condition used by PARC. The oracle true upper bound is a non-deployable diagnostic: it uses audit/latent truth to estimate the best possible release under the same frozen candidates.

Positive-unlabeled baselines. PU-learning methods estimate a classifier or class prior from positives and unlabeled examples. They are relevant because PARC also starts

from verified positives and unverified candidates. However, standard PU risk estimators do not directly output a finite-sample set-level release certificate under block exchangeability and graph compatibility. We therefore report a concrete nnPU classifier-release benchmark and a Bao-style post-selection selective-conformal adaptation as different-target comparators rather than as certified substitutes. The matrix spans the CTC learned, materials and iWildCam human-audit rows and records whether each row has a PARC-style set-level guarantee; the nnPU and Bao-style rows do not. The main Article panel is drawn from the paper-facing frontier CSV (`figure_table2b_baseline_frontier.csv`), the full benchmark summary (`table_pu_selective_conformal_benchmark.csv`) and seed-level rows (`table_pu_selective_conformal_benchmark_seed_rows.csv`). A minimal baseline table (`table_pu_selective_conformal_minimal_baselines.csv`) and the generation script (`scripts/build_pu_selective_conformal_benchmark.py`) are included in the release artifact; the script fixes $\alpha = 0.10$, $K = 100$, $\rho = 0.10$ and 20 fixed split seeds for the comparator benchmark.

Table 10: Comparator target objects. Empirical filters can release individual candidates but do not provide a finite set-level release/refusal certificate under one-sided partial verification. Oracle diagnostics use labels unavailable at deployment.

Method	CTC	Materials	iWildCam
Raw top- K	no certificate	no certificate	no certificate
nnPU	no certificate	no certificate	no certificate
Bao selective conformal	no certificate	no certificate	no certificate
Oracle diagnostic	oracle-only	oracle-only	oracle-only
PARC	certified release	certified release	certified refusal

Why threshold baselines have high conservative diagnostics. Thresholding rewards high detector confidence, and real unannotated objects often have high detector confidence. Under partial verification, unsupported or uncertain released candidates therefore accumulate in the conservative label-uncertainty diagnostic even when the candidate may be real. PARC differs by attaching a set-level certificate to the release decision and by refusing when the null-superset calibration evidence is insufficient.

Truth-known actual-FTR validation. The Article reports two known-truth validation blocks. In the controlled simulation, every candidate unit has latent truth and PARC has no seed-level FTR-bound violation across $\alpha \in \{0.05, 0.10, 0.20\}$; the mean actual FTR at $\alpha = 0.20$ is 2.18×10^{-4} . In the harder adversarial score-overlap simulation, false-candidate scores are shifted toward the verified-positive distribution. Mean actual FTR remains below the target risk at all three risk levels: 0.018 at $\alpha = 0.05$, 0.046 at $\alpha = 0.10$, and 0.111 at $\alpha = 0.20$. The hardest $\alpha = 0.05$ block has a 1% seed-level exceedance, which is compatible with an expectation-level theorem and is reported rather than hidden.

How to read the baseline table. The column “Certificate” is intentionally strict. A method can be calibrated, useful, or visually strong and still lack the PARC false-release certificate. The comparison is therefore not a claim that all baselines are poor trackers; it is a claim that their release decision does not carry the same set-level risk statement under one-sided partial annotations.

Table 11: Truth-known actual-FTR validation summary. The theorem controls expected released-set risk, so the mean actual FTR is the primary validation target; seed-level exceedances are reported separately.

Validation block	α	Trials	Mean actual FTR	Violation rate
Controlled known truth	0.05	100	0.000	0.00
Controlled known truth	0.10	100	0.000	0.00
Controlled known truth	0.20	100	0.000218	0.00
Adversarial score overlap	0.05	100	0.018	0.01
Adversarial score overlap	0.10	100	0.046	0.00
Adversarial score overlap	0.20	100	0.111	0.00

Table 12: Aggregate baseline summary used in the main risk–utility figure.

Method family	Mean release	Conservative FTR diagnostic
Raw top- K / threshold / split conformal	150.0	0.589
Post-filter e-value threshold	58.4	0.174
e-BH-style selection	68.4	0.156
Full PARC	58.4	0.020
Oracle true upper bound	58.8	0.000

Table 13: Baseline taxonomy. “Certificate” means the PARC false-release certificate under the stated assumptions, not merely a calibrated threshold.

Method	Calib.	e-values	SCS	Graph compat.	Certificate
Raw top- K	no	no	no	no	no
Fixed threshold	no/tune	no	no	no	no
Per-generator threshold	yes	no	no	no	no
Split conformal threshold	yes	no	no	no	no FTR certificate
Post-filter e-value	yes	yes	no	no	no SCS certificate
e-BH-style selection	yes	yes	partial	no	not PARC
Full PARC	yes	yes	yes	yes	yes
Oracle upper bound	labels	no	no	optional	diagnostic only

Table 14: Relation to conformal risk control, e-value selection and PARC.

Family	Object controlled	Partial labels?	Graph-compatible set?	PARC distinction
CRC / learn-then-test	Expected risk of a calibrated decision rule.	Usually requires risk labels or observable losses.	Usually not built around domain conflicts.	PARC constructs false-candidate evidence from a one-sided null superset and controls released-set false-release fraction.
PU learning	Classification or class-prior estimation from positives and unlabeled examples.	Yes, but usually as a learning problem.	No set-level compatibility certificate by default.	PARC is a post-hoc release certificate, not a retrained PU classifier.
e-BH / e-FDR	Discoveries among hypotheses using valid e-values.	Requires valid e-values; partial-label construction is external.	No domain graph by default.	PARC adds block-level null-superset e-values, compatibility and the final $K/(\alpha \mathcal{R})$ self-consistency denominator.
PARC	Expected false-release fraction among released candidates.	Yes: verified positives are one-sided and unverified candidates remain in the null superset.	Yes, when the domain exports conflicts.	Certified release or certified refusal in a fixed proposal universe.

Supplement D: Ablation details

The ablation suite separates structural components of the certification layer from diagnostic or stress modifications. Rows marked as actual frozen matrices are produced by rerunning the certification rule on the frozen candidate/e-value matrices. Rows marked as derived sensitivities use the same frozen matrices and alter a documented component such as empty-block interpretation, cell aggregation or label treatment. Negative controls intentionally violate a design principle to expose failure modes; they are not proposed deployment protocols.

Table 15: Ablation summary used for the main-text component discussion. Conservative FTR values are label-uncertainty diagnostics.

Row	Mean release	Cons. FTR	Status
Full PARC	58.4	0.020	actual frozen matrix
Without verified-positive removal	58.4	0.020	actual frozen matrix
Post-filter only, without SCS	58.4	0.174	actual frozen matrix
Conservative empty-block policy	32.1	0.008	derived sensitivity
Coverage-conditional empty-block policy	58.4	0.008	derived sensitivity
Global calibration only	52.6	0.009	derived sensitivity
Random audit-positive removal	43.8	0.010	negative-control sensitivity
Wrongly removing uncertain cases	10.0	0.020	negative-control sensitivity

The no-audit and full-PARC rows can coincide in fixed high-evidence settings. This does not mean audit labels are irrelevant: they establish the one-sided annotation model and are required for safe null-superset shrinkage when the release frontier is sensitive to verified positives. We therefore added a candidate-level rerun on the scientific CTC/materials target rows rather than inferring component behavior from summary tables. Across six target rows and 360 seed-level reruns, leaving verified positives inside the calibration null or removing the same number of random positives reduces release in every row. The effect is a power and release-frontier effect, not an FTR comparison at fixed release size: ablated rows often release fewer candidates and can therefore have lower realized FTR while failing to support the requested certified release.

Table 16: Candidate-level verified-positive removal rerun on CTC/materials target rows. Values are mean released candidates over 20 calibration-mask seeds. The ALIGNN margin-excluded row is retained as a boundary-sensitivity row, not a strict pass.

Target row	Full PARC	No removal	Random removal	Full FTR	Interpretation
CTC learned $\alpha = 0.10, K = 100$	100.00	0.00	0.00	0.000	load-bearing
CTC learned $\alpha = 0.10, K = 300$	300.00	0.00	0.00	0.000	load-bearing
ALIGNN exact-stable $\alpha = 0.10, K = 300$	115.10	0.00	0.00	0.087	load-bearing
ALIGNN exact-stable $\alpha = 0.10, K = 500$	90.75	0.00	0.00	0.048	load-bearing
ALIGNN margin-excluded 25 meV $\alpha = 0.10, K = 100$	100.00	18.40	18.40	0.111	boundary sensitivity
CGCNN exact-stable $\alpha = 0.10, K = 100$	100.00	36.75	36.75	0.030	load-bearing calibration check

Verified-positive removal versus release count. Release count is not the only role of the audit. In the fixed OVT-B rows, the no-verified-positive-removal ablation and Full PARC can return the same mean release because the top evidence mass already clears the self-consistency threshold. This should not be read as evidence that verified positives are unnecessary. The audit supplies the one-sided label model, documents null contamination and enables safe null-superset shrinkage in regimes where the frontier is tighter.

Table 17: Ablation hypotheses and expected failure modes.

Ablation	Question tested	Expected interpretation
Without verified-positive removal	Does audit-based null shrinkage change release power in this fixed matrix?	May coincide with Full PARC when high-evidence mass dominates; still removes a formal component.
Without SCS / post-filter only	Is individual evidence thresholding enough?	Release count can match PARC, but conservative diagnostic increases because set-level denominator is ignored.
Conservative empty-block policy	What is the marginal-valid fallback cost?	More cautious and often lower power.
Coverage-conditional policy	What is the paper-facing covered-regime result?	Practical power under the conditional exchangeability assumption.
Global calibration only	Does coarser calibration help power?	Can increase coverage but may obscure heterogeneity.
Fine Mondrian cells	Does finer conditioning improve diagnostics?	Can reduce unsupported diagnostics but often loses power due to sparse cells.
Random audit-positive removal	Is any removal of calibration paths sufficient?	Negative control; removal must be one-sided reliable.
Wrongly removing uncertain cases	What happens if ambiguous paths are treated as positives?	Negative control; can damage validity/power interpretation.

SCS versus post-filtering. The post-filter e-value baseline demonstrates why set-level self-consistency is not cosmetic. Individual e-values are valid evidence objects for false paths, but the selected set still needs a denominator-aware condition of the form $E_p \geq K/(\alpha|\mathcal{R}|)$. Without this condition, the release may have similar utility but a much larger conservative label-uncertainty diagnostic.

Cell granularity and power. Fine Mondrian cells are not automatically better. They can reduce within-cell heterogeneity and lower unsupported diagnostics, but they also reduce calibration coverage. The OVT-B and TAO cell-granularity rows therefore support the deployed choice of global or fallback calibration for paper-facing power, with fine cells retained as a conservative/power tradeoff diagnostic.

Supplementary Data pointer. The CSV file `table_ablation_components.csv` records dataset, generator, risk level, seed, release count, UTR, audited FTR, conservative FTR, mass ratio, maximum e-value, empty reason, experiment basis, component name, baseline family and notes. This table is the authoritative source for per-row ablation values; the compact table above only summarizes the manuscript-facing contrasts.

Supplement E: Stress tests

Stress rows are reported with provenance sidecars that mark whether they are actual frozen-matrix reruns, actual stress reruns, derived sensitivities, or design rows. Null-inflation stress increases the number of true-but-unverified paths retained in the null superset. Non-exchangeability stress changes calibration/test score distributions. Audit-noise stress perturbs verified-positive flags. Score-miscalibration stress applies monotone and noisy transformations to candidate scores. The main conclusion is qualitative: low-evidence or shifted regimes lead to refusal rather than forced nominal release, while high-evidence regimes remain releasable until the self-consistency mass drops below one.

Table 18: Stress-test provenance summary. Full per-row CSVs are distributed as Supplementary Data.

Stress family	Rows	Main variables	Provenance type
Null inflation	1,800	label keep rate, uncertain inflation	actual rerun + derived label sensitivity
Non-exchangeability	102	dataset/class/size/track shifts	12 actual reruns, 90 derived sensitivities
Audit noise	750	flip/add/remove verified-positive flags	derived audit-noise sensitivity
Score miscalibration	see data	monotone/noisy score transforms	derived sensitivity

E.1 Null-inflation stress

Null inflation simulates the central difficulty of partial annotations: the null superset contains many true but unverified paths. The stress varies the verified-positive keep/removal rate and the uncertain-rate inflation parameter. Rows with lower verified-positive removal rates retain more true positives in the calibration null, which raises block maxima and reduces evidence mass. Rows with higher uncertain inflation increase the conservative label-uncertainty numerator after release. The source CSV `table_stress_null_inflation.csv` records label keep rate, verified-positive removal rate, uncertain-rate inflation, release count, UTR, audited FTR, conservative FTR, mass ratio, e-value ceiling and empty reason.

Table 19: Null-inflation variables.

Variable	Meaning	Source
Label keep rate	Fraction of originally available verified-positive information retained. Lower values make the null superset more inflated.	mixed
Verified-positive removal rate	Fraction of calibration paths removed as verified positives.	rerun
Uncertain-rate inflation	Additional conservative unknown/uncertain mass counted in the diagnostic.	derived
Mass ratio	$\max_k \alpha k E_{(k)} / M$, the uniform self-consistency frontier before conflicts.	computed

E.2 Non-exchangeability stress

Non-exchangeability stress probes how release changes when calibration and test regimes differ. The scenarios include dataset transfer (OVT-B to TAO, TAO to BURST), scene and object shifts (clear to occluded, large to small, long to short), category-frequency shift (head to tail), and severe sparse-annotation shift. Some rows are actual reruns of the certification matrix under a changed

split; others are derived sensitivities computed from frozen source matrices. The provenance sidecar marks this distinction for every row. The main manuscript uses reader-facing labels such as “head $\rightarrow$ tail”; the source CSV retains machine-readable identifiers for reproducibility.

Table 20: Reader-facing labels for non-exchangeability stress.

Manuscript label	Source-field meaning
OVT-B $\rightarrow$ TAO	Calibration distribution from OVT-B, test distribution from TAO-style rows.
TAO $\rightarrow$ BURST	Calibration distribution from TAO, test distribution from BURST-style rows.
clear $\rightarrow$ occluded	Scene visibility shift.
head $\rightarrow$ tail	Query/category frequency shift.
large $\rightarrow$ small	Object-scale shift.
long $\rightarrow$ short	Temporal-support shift.
sparse annotation	Severe sparse-annotation stress row.

E.3 Audit-noise stress

Audit-noise stress perturbs verified-positive flags and diagnostic label assignments. False-to-true flips are the dangerous direction for null-superset validity because they may remove a false path from calibration. True-to-false flips are conservative for validity but can reduce power by keeping more true paths in the null superset. The manuscript treats these rows as assumption-boundary diagnostics rather than new benchmark reruns. The source CSV `table_stress_audit_noise.csv` records the noise type, noise rate, release count, conservative FTR and empty reason.

E.4 Score-miscalibration stress

Score-miscalibration stress applies monotone or noisy transformations to path scores. Pure monotone transformations preserve score ranks and therefore often preserve rank-based release behaviour when the same transformation is applied consistently to calibration and test. Noisy or split-dependent transformations can change block maxima, evidence mass and graph selection. PARC is not a replacement for a stable proposal score: it certifies release under the frozen scoring rule and exchangeability assumptions supplied to calibration.

Interpretation boundary. Stress tests should be read as assumption-boundary diagnostics. They do not create new benchmark claims and they do not replace the formal theorem. Their purpose is to show how the certification layer behaves when the assumptions that make the theorem useful become strained: release shrinks, diagnostics worsen, or the system refuses.

E.5 Block-size heterogeneity robustness

The block-size heterogeneity package separates completed candidate-level reruns from scoped aggregate diagnostics. For materials screening, the public WBM candidate table is available in the derived package, so the certification matrix can be rerun after size-matching calibration blocks and after downsampling block maxima. For CTC and SpaceNet, the review package does not redistribute the candidate-level artifacts required for an equivalent rerun; those rows are therefore labelled as aggregate-only or audit-sample diagnostics and are not used as completed candidate-level

robustness evidence. This scope distinction prevents the robustness table from silently converting unavailable candidate-level reruns into paper claims.

Table 21: Block-size heterogeneity robustness status. Materials rows are completed candidate-level diagnostics; CTC and SpaceNet rows are scoped diagnostics because the public package does not contain the necessary candidate-level artifacts.

Domain	Completed diagnostic	Scope status	Interpretation
Materials screening	size-stratified p-value diagnostic; 12 size-matched rerun rows; 8 downsampled block-max rows	candidate-level completed	Release/refusal pattern is retained or power becomes conservative under stricter block-size matching.
Biomedical cell tracking	aggregate block-size summary	aggregate-only public package	No candidate-level rerun is claimed without redistributed CTC candidate artifacts.
Earth observation	audit-sample block summary and aggregate diagnostic	audit-sample diagnostic	The audit sample is underpowered for a candidate-level rerun and is reported only as a scoped diagnostic.

The source files are `table_block_size_heterogeneity_summary.csv`, `figure_block_size_superuniformity.csv`, `table_size_matched_rerun.csv` and `table_downsampled_blockmax_stress.csv`. The figure file `extended_data_block_size_superuniformity.pdf` visualizes the size-stratified diagnostic and should be read as an assumption check rather than as a new theorem.

Supplement F: Human audit protocol

Audited paths are labelled as actually true, actually false or uncertain. A stricter verified-positive flag is assigned only when the object identity, path localization and semantic query match are clear enough for calibration use. Uncertain cases are treated conservatively in reported diagnostics. The BURST audit follows the same two-pass author visual-review protocol as OVT-B and TAO; no model-generated labels are used to assign BURST verified-positive status. These open-world audits are reported as annotation-workflow evidence and reproducibility support. They motivate the paper’s null-superset treatment of unsupported benchmark candidates, but they are not the primary FTR evaluation source for the CTC or SpaceNet 7 scientific-domain rows. In the Audit2000 label-count summary, high-scoring unsupported rows are often visually true, with 1,927 actually true, 33 actually false and 40 uncertain rows under the final taxonomy; this observation is used as evidence of annotation incompleteness rather than as a cross-domain theorem validation.

F.1 Label taxonomy

Table 22: Audit label taxonomy.

Audit label	Visual criterion	Diagnostic treatment
Actually true	A real object instance matching the query/category is visible and the path localization is acceptable for the audited temporal extent.	Not counted as false in audited FTR.
Actually false	The path is a hallucination, background region, wrong semantic object, severe drift, or otherwise not a valid tracklet.	Counted as false.
Uncertain	Visual evidence is ambiguous due to occlusion, heavy truncation, category ambiguity, severe localization uncertainty or insufficient temporal support.	Counted conservatively in conservative FTR.
Verified positive	A stricter subset of actually true: semantic match, localization and temporal consistency are clear enough to remove the path from the calibration null.	May be removed from calibration null.

F.2 Review information and blindness

Reviewers inspect candidate-path visual evidence and metadata needed to apply the taxonomy: query/category name, path identifier, sampled frames or montage crops, and the path’s localization over time. The closeout second review is blind to the prior round’s assigned label and verified-positive flag. The second review is not instructed by model-generated labels. Score and source metadata can be present as provenance, but the label guideline instructs reviewers to decide from visual evidence and the semantic query rather than from score rank. When visual evidence is insufficient, the uncertain label is used.

For the closeout reliability check, the paper-facing package separates hard-case ambiguity measurement from primary release-evaluation evidence. The boundary challenge contains 500 deliberately difficult rows sampled from previous uncertain/false neighborhoods, low- or medium-confidence rows, small objects, partial boxes, category-boundary cases, temporal fragments and released-unsupported boundary examples. It yields 25 disagreements, label agreement 0.950 and Cohen’s $\kappa = 0.833$, with exact verified-positive agreement. The full Audit2000 blind reannotation yields 25 disagreements over 2,000 rows, label agreement 0.9875 and Cohen’s $\kappa = 0.849$, again with

Table 23: Blind-review fields.

Field	Visible in closeout review?	Purpose
Path montage / sampled crops	yes	Primary visual evidence.
Text query or category	yes	Determines semantic match.
Path id and dataset id	yes	Provenance and bookkeeping.
Previous audit label	no	Prevents direct copying of first-pass label.
Previous verified-positive flag	no	Prevents direct copying of calibration-grade flag.
Model-generated label	no	No model-generated labels assign verified-positive status.

Table 24: Annotator independence and visibility. “Independent” here means blind to the previous label within the project audit workflow; it does not imply an external commercial annotation vendor.

Item	First review	Closeout second review
Reviewer role	Project-team first visual audit	Project-team blind visual review
Previous label visible	no	no
Verified-positive flag visible	no	no
Model-generated label visible	no	no
Release decision visible	no	no
Query/category visible	yes	yes
Path montage or sampled crops visible	yes	yes
Score/source metadata	provenance only	provenance only
Uncertain label allowed	yes	yes

exact verified-positive agreement. These values are used as annotation-workflow credibility checks, not as the main empirical claim of the Article and not as a substitute for CTC or SpaceNet held-out FTR measurement. Agreement is computed as exact agreement on the three-way label and exact agreement on the verified-positive flag. For labels $L_r^{(1)}, L_r^{(2)}$ and verified-positive flags $V_r^{(1)}, V_r^{(2)}$, the reported exact-agreement rates are

$$\frac{1}{N} \sum_{r=1}^N \mathbf{1}[L_r^{(1)} = L_r^{(2)}], \quad \frac{1}{N} \sum_{r=1}^N \mathbf{1}[V_r^{(1)} = V_r^{(2)}].$$

Disagreement policy, if triggered, is consensus review with the uncertain label used unless the visual evidence is clear. The row list, label files and SHA256 provenance are included in the supplementary archive.

The audit closeout is not presented as a replacement for the formal risk theorem. It is an audit-quality check showing that the final taxonomy and verified-positive rule are reproducible, including on deliberately difficult boundary cases.

Table 25: Audit2000 primary label-count summary. These counts motivate the null-superset treatment of unsupported open-world candidates; they are not used as primary CTC or SpaceNet 7 evidence.

Dataset	Rows	Actually true	Actually false	Uncertain	Verified positive
OVT-B	700	655	18	27	38
TAO	700	685	15	0	32
BURST	600	587	0	13	25
Total	2,000	1,927	33	40	95

Table 26: Blind-review agreement under the updated audit closeout. Boundary-500 is the hard-case ambiguity challenge; Audit2000 reannotation is the full benchmark reannotation. Both rows have exact verified-positive agreement.

Review block	Rows	Label agreement	Cohen’s κ	Disagreements	VP agreement
Boundary-500 challenge	500	0.9500	0.8326	25	1.0000
Audit2000 reannotation	2,000	0.9875	0.8490	25	1.0000

F.3 Sampling and agreement interpretation

The boundary challenge is designed to measure ambiguity in hard cases, not to tune agreement to a target value. Its 25 disagreements are all resolved conservatively into uncertain in the second-review labels. The full Audit2000 reannotation gives the same number of disagreements over a larger sample, again by moving boundary true cases into uncertain rather than creating additional false or verified-positive rows. For this reason, the agreement results should be read as audit-workflow reliability checks, not as theorem-level empirical validation of all latent labels.

F.4 Disagreement and adjudication policy

If a second-review row disagrees with the primary audit label, the row is sent to consensus review. The default consensus outcome is uncertain unless the visual evidence is clear. If the disagreement concerns verified-positive status rather than the three-way visual label, the path is not used as a verified positive unless both reviews agree that it satisfies semantic, localization and temporal consistency criteria. This policy is intentionally conservative: disagreement cannot create an additional calibration positive without consensus.

F.5 Audit file provenance

The supplementary archive includes the label-count summaries, Boundary-500 challenge summary, Audit2000 reannotation summary, SHA256 manifests and provenance sidecars identifying the review version. The archive does not redistribute raw benchmark video frames. If montage files contain benchmark imagery, they are treated as restricted review artifacts under the original dataset licenses rather than public data.

Table 27: Second-review label distributions. The full reannotation shifts 25 primary actually-true rows into uncertain, reflecting conservative handling of boundary ambiguity.

Review block	Actually true	Actually false	Uncertain
Boundary-500 primary	427	33	40
Boundary-500 second review	402	33	65
Audit2000 primary	1,927	33	40
Audit2000 second review	1,902	33	65

Table 28: Verified-positive precision margin in reviewed subsets. No false verified-positive was observed among reviewed verified-positive cases. The upper bound is the one-sided exact binomial 95% upper bound for zero observed false verified positives, using the number of reviewed verified-positive cases as denominator.

Reviewed set	Reviewed VP cases	False VP observed	95% upper bound
Audit2000 reannotation	95	0	0.031

Supplement G: Generality

The paper uses separate naming for generality rows. LVIS denotes single-frame open-vocabulary detection. LVVIS or LV-VIS denotes video or mask-path settings. OVVIS/mask-path scaffold rows are proof-of-principle evidence unless full official mask benchmark provenance is available. Main claims about certified tracking release are made on OVT-B, TAO and the box-converted BURST validation protocol; LVIS and mask-path rows are scoped as generality evidence.

Table 29: Task naming and permitted claims.

Name	Task scope	Claim allowed in this manuscript
OVT-B	Open-vocabulary video tracking benchmark.	Main tracking certification evidence.
TAO	Federated video tracking/long-tail annotation regime.	Main hard partial-annotation regime and refusal evidence.
BURST boxes	Box-converted validation protocol from segmentation annotations.	Third annotation-regime validation; not a native OVMOT replacement.
LVIS	Single-frame open-vocabulary detection.	Generality evidence for single-frame release; not tracking.
LVVIS / LV-VIS	Video or mask-path task, depending on source.	Scoped appendix evidence only unless official provenance is complete.
OVVIS / mask-path scaffold	Box-to-mask or video-instance proof-of-principle scaffold.	Proof-of-principle; no full official mask benchmark claim.

LVIS detection rows. LVIS rows are single-frame release-certification evidence. The GroundingDINO rows are positive official-support-proxy rows. OWLv2 rows are retained as stress evidence because high-score unmatched audit labels remain pending. These rows support the idea that the null-superset release logic can be applied beyond video tracking, but they do not establish a full open-vocabulary tracking claim. The LVIS audit scaffold currently contains 500 high-score unmatched candidates, all marked as requiring human review; no LVIS human-audit FTR is claimed in the Article.

Table 30: LVIS audit-closeout status. The scaffold is present, but labels remain open; therefore LVIS rows are reported as official-support-proxy cross-task evidence rather than visual-audit FTR evidence.

Detector	Candidates	Human-labelled	Verified-positive	Status
GroundingDINO	139	0	0	requires human review
OWLv2	361	0	0	requires human review

Mask-path and OVVIS/LVVIS rows. Mask-path rows are treated as proof-of-principle. They may use box-to-mask conversion or mask prompting, and are reported only when the paper can state the conversion scope without implying an official mask benchmark result. If full official mask benchmark provenance is unavailable, the row remains appendix evidence and cannot be used as a headline result.

Published tracker adapters. Published tracker rows are useful because they test whether PARC can wrap existing systems. However, public tracker outputs require careful provenance: official prediction file hashes, conversion scripts, split definitions, category-query mapping and support/matching conventions. Without this provenance, the row demonstrates a failure mode or wrapper diagnostic but is not a leaderboard comparison.

Supplement H: Reproducibility

The supplementary archive contains derived CSV tables, audit summaries, released-set diagnostics, figure-generation inputs, provenance JSON sidecars and SHA256 manifests. The review repository contains certification code, table-generation scripts, public fixtures, environment files, Makefiles and continuous-integration recipes. Raw benchmark videos, raw benchmark annotations, model weights, detector caches and frame caches are not redistributed and remain subject to their original licenses. Publication figures are regenerated with Matplotlib, Seaborn and SciencePlots and exported as vector PDF. OpenCV and supervision are installed for CV montage and detection/track overlay generation when qualitative figures are included.

Core commands are:

```
make test
make tiny-fixture
make reproduce-main-tables
make reproduce-main-figures
make validate-public-bundle
make verify-manifest
```

The public-safe bundle excludes raw datasets, raw annotations, model weights, detector caches and frame caches. SHA256 manifests cover derived tables, figures, audit summaries and provenance sidecars.

Table 31: Reproducibility command map.

Command	Purpose	Requires raw data?
<code>make test</code>	Run unit tests for score construction, label metrics, SCS self-consistency and bundle validation.	no
<code>make tiny-fixture</code>	Run the certification pipeline on a public-safe synthetic fixture.	no
<code>make reproduce-main-tables</code>	Rebuild manuscript tables from derived CSV summaries.	no
<code>make reproduce-main-figures</code>	Regenerate publication figures from figure input CSVs.	no
<code>make validate-public-bundle</code>	Check that public artifacts do not include raw frames, model weights or private caches.	no
<code>make verify-manifest</code>	Verify SHA256 hashes for derived artifacts and provenance sidecars.	no
<code>phase2 propose</code>	Generate raw candidate universes from videos and models.	yes
<code>phase3 matrix</code>	Build certification matrices from candidate universes.	depends on candidate files

Environment. The plotting stack used for publication figures is Matplotlib, Seaborn and SciencePlots. OpenCV and supervision are installed for computer-vision overlay and montage construction. Interactive libraries such as Plotly and Altair are optional and used only for exploration; paper figures are exported as static vector PDF. The repository includes environment and lock-file instructions so that reviewers can rebuild tables and figures without downloading raw benchmark data.

Table 32: Included and excluded artifact classes.

Artifact	Review package	Public release after acceptance
Derived CSV summaries	included	included
Figure input CSVs and vector PDFs	included	included
Audit summaries and second-review counts	included	included where licenses permit
Provenance JSON sidecars and SHA256 manifests	included	included
Certification code and plotting scripts	included in anonymized form	public repository
Raw benchmark videos and annotations	excluded	excluded; original licenses apply
Model weights and HF/cache files	excluded	excluded; users download from original providers
Frame caches and private montage viewers	excluded	excluded unless dataset license allows redistribution

Supplement I: Extended Data and Supplementary Data inventory

The Extended Data package contains publication-facing versions of the following supporting displays: evidence map; full baseline risk–utility frontier; ablation heatmap; null-inflation sensitivity; non-exchangeability shifts; audit-noise and score-miscalibration sensitivity; block-size heterogeneity diagnostics; human-audit sample composition and confusion matrix; LVIS raw detector versus PARC; mask-path proof-of-principle rows; visual release/refusal examples; release-card maps; prospective-protocol status; and a reproducibility/provenance checklist. The Supplementary Data package contains the per-seed CSVs, figure input CSVs, provenance JSON sidecars and SHA256 manifests. Large raw datasets, raw annotations, model weights and detector caches are excluded.

Exact filenames are listed below.

SD0	table_evidence_map.csv
SD1	table_main_raw_vs_parc_summary.csv
SD1b	table_released_set_audit_summary.csv
SD1c	table_main_raw_vs_parc.csv
SD1d	table_alpha_frontier.csv
SD2	table_baseline_comparison_summary.csv
SD3	table_ablation_components.csv
SD4	table_safe_refusal_diagnostics.csv
SD5a	table_stress_null_inflation.csv
SD5b	table_stress_nonexchangeability.csv
SD5c	table_stress_audit_noise.csv
SD5d	table_stress_score_miscalibration.csv
SD6	figure_stratified_reliability.csv
SD7a	audit_labels_2000_human_reviewed_summary.csv
SD7b	audit_labels_2000_human_reviewed_verified_summary.csv
SD7c	audit_boundary_challenge_500_agreement_summary.csv
SD7d	audit2000_reannotation_agreement_summary.csv
SD7e	table_second_review_challenge_closeout.csv
SD7e2	audit_boundary_challenge_500_confirmed_disagreements.csv
SD7e3	audit2000_human_reannotation_labels.csv
SD7f	table_actual_ftr_validation_summary.csv
SD7g	table_actual_ftr_validation.csv
SD7h	table_actual_ftr_hard_regime_summary.csv
SD7i	table_seed_variability_ftr_uncertainty.csv
SD7i2	table_pu_selective_conformal_benchmark.csv
SD7i3	table_pu_selective_conformal_benchmark_seed_rows.csv
SD7i4	figure_table2b_baseline_frontier.csv
SD7i5	PU_SELECTIVE_CONFORMAL_BENCHMARK_CLOSEOUT.md
SD7i6	table_runtime_compute_overhead_scientific_domains.csv
SD7i7	table_iwildcam_second_review_status.csv
SD7i8	table_iwildcam_second_review_agreement_summary.csv
SD7i9	table_iwildcam_second_review_disagreement_cases.csv
SD7i10	iwildcam_second_review_human_confirmed_labels.csv
SD7i11	iwildcam_second_review_human_confirmed_comparison.csv
SD7i12	table_materials_modern_model_sensitivity.csv
SD7i13	table_no_human_consequence_summary.csv
SD7i14	figure_no_human_consequence_main.csv
SD7i15	figure_materials_model_zoo_frontier.csv
SD7i16	NO_HUMAN_PAPER_INTEGRATION.md
SD7i17	table_materials_computational_trial_summary.csv
SD7i18	figure_materials_computational_trial_main.csv
SD7i19	table_official_downstream_consequence_summary.csv
SD7i20	figure_official_downstream_consequence.csv
SD7i21	table_release_certification_track_registry.csv
SD7i22	table_release_certification_cards.csv
SD7i23	table_release_certification_benchmark_index.csv
SD7i24	table_release_card_field_schema.csv
SD7i25	RELEASE_CERTIFICATION_GOVERNANCE_PROTOCOL.md
SD7i26	table_verified_positive_removal_load_bearing.csv
SD7i27	table_verified_positive_removal_load_bearing_seed_rows.csv
SD7i28	table_block_size_heterogeneity_summary.csv
SD7i29	figure_block_size_superuniformity.csv

SD7i30 table_size_matched_rerun.csv
SD7i31 table_size_matched_rerun_seed_rows.csv
SD7i32 table_downsampled_blockmax_stress.csv
SD7i33 table_downsampled_blockmax_stress_seed_rows.csv
SD7i34 B2_APPROXIMATE_EVALUATE_VALIDITY_LEMMA.md
SD7i35 table_materials_prospective_validation_go_no_go.csv
SD7i36 table_materials_temporal_split_feasibility.csv
SD7i37 table_materials_independent_dft_source_feasibility.csv
SD7i38 table_materials_prospective_validation_cards.csv
SD7i39 table_dft_followup_freeze_status.csv
SD7i40 table_unlabeled_pool_build_status.csv
SD7i41 table_selection_freeze_status.csv
SD7i42 table_dft_job_export_status.csv
SD7i43 table_alignnff_score_status.csv
SD7i44 MATERIALS_TEMPORAL_REPLAY_CLOSEOUT.md
SD7i45 table_temporal_claim_scope.csv
SD7i46 table_temporal_primary.csv
SD7i47 table_temporal_raw_vs_parcs.csv
SD7i48 table_temporal_snapshot_inventory.csv
SD7i49 FIXED_BUDGET_SCIENTIFIC_UTILITY_CLOSEOUT.md
SD7i50 table_fixed_budget_primary.csv
SD7i51 table_fixed_budget_seed_rows.csv
SD7i52 table_decision_curve.csv
SD7i53 table_false_followups_prevented.csv
SD7i54 table_cost_per_true_candidate.csv
SD7i55 table_refusal_value.csv
SD7i56 table_ctc_downstream_artifact_utility.csv
SD7i57 table_spacenet_downstream_artifact_utility.csv
SD7i58 ADVERSARIAL_RELEASE_STRESS_CLOSEOUT.md
SD7i59 table_adversarial_stress_trials.csv
SD7i60 table_refusal_boundary.csv
SD7i61 table_stress_claim_scope.csv
SD7i62 SELECTOR_OPTIMALITY_DIAGNOSTICS_CLOSEOUT.md
SD7i63 table_greedy_vs_ilp.csv
SD7i64 table_mass_vs_graph_failure.csv
SD7i65 table_conflict_loss.csv
SD7i66 table_selector_claim_scope.csv
SD7j spacenet7_real_audit_closeout.json
SD7k table_spacenet7_real_audit_k100_failure_summary.csv
SD7l table_spacenet7_real_audit_k50_completed_summary.csv
SD7m spacenet7_real_audit_human_gate.json
SD7n table_spacenet7_real_audit_calibration_summary.csv
SD7o table_spacenet7_real_audit_block_coverage.csv
SD7p table_spacenet7_real_audit_block_coverage_summary.csv
SD7q table_spacenet7_real_audit_primary_refusal_diagnostics.csv
SD7r table_spacenet7_real_audit_k100_evaluate_failure_by_seed.csv
SD7s table_spacenet7_real_audit_raw_topK_audit.csv
SD7t table_spacenet7_real_audit_seed_results.csv
SD7u table_spacenet7_real_audit_k50_release_audit.csv
SD8 table_lvis_detection_main.csv
SD8b table_lvis_raw_detector_vs_parcs.csv
SD8c audit_labels_lvis_status_summary.csv
SD9 table_mask_path_proof_of_principle.csv

Table 33: Extended Data display inventory.

Display	Content	Source
Extended Data Fig. 1	Main protocol coverage map over datasets, proposal sources, seeds and inclusion status.	main protocol coverage CSV
Extended Data Fig. 2	Full baseline risk-utility frontier including all threshold baselines.	baseline comparison CSV
Extended Data Fig. 3	Ablation component heatmap and component taxonomy.	ablation component CSV
Extended Data Fig. 4	Null-inflation sensitivity over label keep rate and uncertain inflation.	null-inflation CSV
Extended Data Fig. 5	Non-exchangeability shifts with actual-rerun versus derived-sensitivity markers.	non-exchangeability CSV
Extended Data Fig. 6	Audit-noise and score-miscalibration sensitivity.	audit-noise and score-miscalibration CSVs
Extended Data Fig. 7	Human-audit sample composition and confusion matrix.	audit summary and second-review CSVs
Extended Data Fig. 8	LVIS detection raw detector versus PARC scoped evidence.	LVIS detection CSV
Extended Data Fig. 9	Mask-path proof-of-principle rows.	mask-path proof-of-principle CSV
Extended Data Fig. 10	Visual examples of released and refused candidate units across microscopy, satellite and camera-trap workflows.	qualitative asset bundle
Extended Data Fig. 11	Success-domain map summarizing evidence mass, coverage, raw risk and release/refusal outcomes.	success-domain feature CSVs
Extended Data Fig. 12	Release-certification benchmark-card map over reusable tracks and downstream artifacts.	release-certification benchmark CSVs
Extended Data Fig. 13	Block-size heterogeneity and size-stratified super-uniformity diagnostics.	block heterogeneity robustness CSVs
Extended Data Table 1	Evidence map summarizing primary, robustness, audit, boundary and breadth evidence classes.	evidence map CSV
Extended Data Table 2	Reproducibility and provenance checklist.	SHA256 manifests and reproducibility scripts
Extended Data Table 3	Release-certification benchmark tracks and card schema.	release-card registry CSVs
Extended Data Table 4	Prospective-materials protocol and feasibility status for A1/A2/A3.	prospective validation status CSVs

Table 34: Extended Data Table 3: release-certification benchmark tracks. Each track is represented by release cards in the supplementary archive; the table summarizes the reusable governance object rather than adding a new experimental claim.

Track	Candidate unit	Verification source	Downstream artifact	Decision type
Materials follow-up	crystal candidate	public DFT labels	computational follow-up queue	release / refusal
CTC cell-link release	adjacent-frame cell link	masked official positives; queue audit	cell lineage graph	strict release
iWildCam ecology	animal-present box	human audit	animal-detection release list	operational release
CTC lineage guardrail	adjacent-frame cell link	official CTC identities	cell lineage graph	refusal / guardrail
SpaceNet map guardrail	same-building temporal link	official building identities	building-persistence map	refusal / guardrail

Table 35: Extended Data Table 4: prospective materials validation protocol status. These rows are preregistered or feasibility-scoped protocols and are not completed evidence in the present manuscript.

Protocol	Intended validation	Current blocker	Manuscript status
A1 temporal split	Evaluate release against a future or timestamped public-label snapshot	External timestamp or benchmark-release snapshot required	protocol / feasibility only
A2 independent DFT cross-validation	Join released candidates to an independent DFT label source	External source mapping and join table required	protocol / feasibility only
A3 prospective DFT follow-up	Freeze ALIGNN-FF $\alpha = 0.10, p = 0.10, K = 500$ follow-up arms	Missing unlabeled generated candidate pool; frozen selection and job manifest are empty	protocol frozen, blocked; no DFT results

Table 36: Evidence map for the release-certification package. This table makes the supporting experiment matrix visible without turning every robustness or boundary row into a primary main-text claim. Artifact paths are paper-facing public-bundle locations or the corresponding exported CSV names.

Evidence class	Domain/source	Scale or result	Role	Placement	Artifact
Flagship positive	CTC Internal Split	$\mu = 0.10, \alpha = 0.10, K = 10-300$; 20/20 non-empty; hold-out FTR 0.000	primary positive	Table 1 / Fig. 5	CTC Internal split CSV
Flagship positive	Materiality Dictionary CCGSN	WBM unique prototype; $\mu = 0.10, \alpha = 0.10, K = 100$; 20/20 non-empty; hold-out FTR 0.000	predefined calibration check	Table 1 / Fig. 5	materiality primary CSV
Split validation	CTC reverse split	Train sequence 12 = error sequence 10; 20/20 calibration-mask mode; FTR 0.000	robustness	Article text / Supp. C	CTC reverse split CSV
Leakage audit	CTC Internal source	No train-test sequence overlap; no GT identity or matching-label fields; train-only normalization	source validity	Article text / Supp. C	CTC leakage audit CSV
Negative control	CTC random score	Raw top-K FTR 0.36-6.41; all FABC rows refused	source-quality check	Fig. 2 / Supplement	CTC random-score CSV
Negative control	Materiality random score	Raw top-K FTR about 0.65; FABC rows refused	source-quality check	Fig. 3 / Supplement	materiality random-score CSV
Model identity	Materiality ALIGNN-FF	$K = 300$; 100% verified dropout release 115/111 candidates with FTR 0.007/0.006; full row top-K response base FTR 0.253/0.327	requested budget value demonstration, not fixed-size rank improvement	Fig. 2 / Article text	materiality modern-model CSV
Scientific consequence	Materiality computed-label follow-up	ALIGNN-FF top-500 presents 154.3 suitable follow-ups per seed; top-5000 refused presents 2,377.4 suitable follow-ups per seed	downstream impact translation	Article text / Supplement	materiality computational trial CSVs
Fixed-budget utility	Materiality model run	CCGSN/ALIGNN-FF/MEGNet $K = 300$ rows present 158.25/158.31/165.4 suitable follow-ups per seed; $K = 3000$ release presents 2,032.6/2,077.4/2,083.5	completed public-label utility evidence	Article text / Supplementary Data	fixed-budget scientific utility CSVs
Scientific consequence	Official downstream artifacts	CTC using $K = 5000$ presents 2,807.5 false-leasing edges per seed; CTC random-score presents 3,097.5; SpacNet randomized presents 3,309.9 false persistence links	official downstream consequence CSVs	Article text / Supplement	official downstream consequence CSVs
Biobank audit	Release-certification checks	Free results tracks covering materials follow-up, CTC link release, (WBM)Can annual release, CTC leakage guardrail and SpacNet persistence-map guardrail	off-to-label consequence metrics	Extended Data / Data package	release-certification benchmark CSVs
Black integrity	Materiality / CTC / SpacNet	Materiality candidate-level one-matched and downstream events; CTC aggregate-only; SpacNet multi-sample diagnostic	genomewide provenance	Supp. E / Extended Data	black integrity consequence CSVs
Adversarial stress	Release release rows	Stress corruption, sample high-K response and black-shield/downsampled stress rows absent or refuse unsupported release	completed diagnostic	Supp. E / Supplementary Data	adversarial release stress CSVs
Selector optimality	Refused rows	Classify vs CLP diagnostic check reported referrals on finite-resolution or pre-graph issue failures, not grossly computability artifacts	completed diagnostic	Supp. E / Supplementary Data	selector optimality CSV
Prospective protocols	Materiality validation A1/A2/A3	Tempered reply results as available $\frac{1}{2}$; sequential independent DFT sends a join table; prospective DFT follow-up is frozen but blocked by external inputs	protocol / feasibility only	Supp. E / Supplementary Data	prospective validation protocol CSVs
Component addition	Verified-positive removal	0 CTC/materiality target rows; 300 seed-level events; no removal or random-removal release volume in 0/0 rows	load-bearing component check	Extended Data / Data package	verified-positive-removal load-bearing CSV
Leak-validation check	SpacNet T geometry	10 MoA, 3,348 links; $\alpha = 0.20$; 17/20 non-empty, below the predefined non-empty gate; hold-out FTR 0.003	release model-label operating check	Article text / Supp. D	SpacNet geometry CSV
Real audit	SpacNet T human audit	800 calibration candidates; $K = 100$ released; $K = 10, \alpha = 1$ LF release audit; human FTR 0.000	deployment check	Table 1 / Article text	SpacNet audit CSV
Human-audited positive	(WBM)Can annual-present	$\alpha = 0.20, K = 10$; 20/20 non-empty; 167/167 release audit; human FTR 0.000	operational ecology release	Table 1 / Fig. 4	(WBM)Can human-audit CSV
Unaided events	SpacNet randomized	Raw top-K FTR about 0.66-6.66; FABC rows refused	safety refusal	Supplement	SpacNet randomized CSV
Boundary	(WBM)Can spectra prompts	Strata $\alpha = 0.10$ pre-graph boundary	assumption boundary	Table 1 / Fig. 4	(WBM)Can assumption CSV
Boundary	(WBM)Can strict material-present	Training, path and detection rows outside primary scientific anchors	strict-risk boundary	Table 1 / Fig. 4	(WBM)Can strict-boundary CSV
Diagnosics	OFT-4/BEISF/TMG-LVS	Tracking, path and detection rows outside primary scientific anchors	auxiliary granularity	Supp. E / ED	granularity_reliability CSV
Diagnosics	Black, seed and provided-release rows	Recomputability, variability and refusal diagnostic	assumption support	Supplementary Data	release_attr/paper_alignn/
Baseline	mPU/selective confoundal benchmark	CTC materials and (WBM)Can rows; no FABC set-level guarantee for comparator rows	different-target baseline	Supp. D	mPU/selective benchmark CSV
Baseline	Scientific-domain certification overhead	WBM table-level grid about 4.5 x excluding model inference; SpacNet selective adversarial after precomputed black ids	complete diagnostic	Supplementary Data	runtime overhead CSV
Second review	(WBM)Can human audit	1,123 rows; 149 disagreements; label agreement 0.902; $\alpha = 0.804$ [0.708, 0.838]; release release 167/167	audit reliability	Supp. F	(WBM)Can second-review CSVs

Table 37: Supplementary Data index.

File	Purpose
table evidence map	Primary, robustness, audit, boundary, breadth and diagnostics evidence matrix.
table main raw-vs-PARC summary	Main fixed-protocol release/refusal and raw top- K comparison.
table released-set audit summary	Released-set support/audit bridge used by the main quantitative table.
table main raw-vs-PARC full	Per-seed released-set support/audit fields and evidence diagnostics.
table alpha frontier	Risk-level sensitivity rows, including the TAO $\alpha = 0.20$ positive operating point.
table baseline comparison summary	Aggregate baseline comparison used in risk-utility analysis.
table ablation components	Component ablations and derived sensitivities.
table safe-refusal diagnostics	Evidence-mass and empty-reason rows for refusal diagnostics.
table stress null inflation	Null-inflation stress inputs.
table stress non-exchangeability	Dataset, covariate and annotation-shift stress inputs.
table stress audit noise	Audit-noise sensitivity inputs.
table stress score miscalibration	Score-miscalibration sensitivity inputs.
figure stratified reliability	Stratified reliability heatmap inputs.
audit labels 2000 summary	Audit2000 label-count summary.
boundary challenge agreement summary	Boundary-500 hard-case agreement and kappa summary.
Audit2000 reannotation agreement summary	Full Audit2000 blind reannotation agreement and kappa summary.
table second-review challenge close-out	Paper-facing boundary and Audit2000 closeout rows.
table actual-FTR validation summary	Known-truth simulation and hard-regime actual-FTR validation summary.
table actual-FTR validation	Per-seed known-truth actual-FTR validation rows.
table seed variability FTR uncertainty	Seed-level release variability and seed-bootstrap FTR intervals for primary rows.
table block exchangeability diagnostics main	Compact main-text block coverage and assumption-boundary diagnostics.
table CTC learned-hybrid main	Sequence-disjoint learned-hybrid CTC release rows and strict $\alpha = 0.10$ small- K extension.
table CTC learned reverse split	Reverse sequence-disjoint split rows for train sequence 02 and held-out sequence 01 certification.
table CTC learned negative control	Random-score negative-control rows showing high raw false-link rates and certified refusal.
table CTC learned leakage audit	Feature, split and normalization audit for the learned-hybrid source and controls.
table CTC learned model report	Learned-hybrid source provenance, features and leakage-exclusion summary.
table CTC learned versus geometry	Structured-source comparison between learned-hybrid and geometric CTC rows.
table materials primary results	Matbench Discovery/WBM CGCNN release rows, including the strict $\alpha = 0.10, K = 100$ primary endpoint and sensitivity rows.
table materials candidate universe summary	WBM unique-prototype universe size, DFT-stable count and block definition.

Supplement J: Extended Data captions and construction notes

This note records the construction of the Extended Data displays. The main Article keeps six display items; the Extended Data package carries broader evidence without overloading the main narrative. Each Extended Data display is generated from the Supplementary Data files listed in Supplement I and is read with the same scope limitations as the main text.

Extended Data Table 1: Evidence map. The evidence map is a scope-control device rather than a new result. It summarizes which rows are primary positives, which rows are robustness checks, which rows are human-audit operating checks, which rows are boundary diagnostics and which rows are auxiliary breadth evidence. This prevents the main Article from reading as a benchmark pile while making the full certification package auditable.

Extended Data Fig. 1: Main protocol coverage map. This display visualizes which dataset-generator-risk-level rows are included in the main protocol and which rows are appendix, stress or scoped generality evidence. Rows are grouped by dataset (OVT-B, TAO, BURST, LVIS and mask-path scaffolds), proposal source (GroundingDINO, GroundingDINO plus tracker, OWLv2, OWL-ViT and published tracker adapters), and risk level. The figure distinguishes complete three-seed rows from incomplete or provenance-only rows. Its purpose is not to add a new result; it is a reader map for the experiment scope.

Extended Data Fig. 2: Full baseline risk-utility frontier. The main risk-utility figure shows the representative frontier. The Extended Data version includes raw top- K , fixed score threshold, per-generator calibrated score threshold, split-conformal p-value threshold, post-filter e-value threshold, e-BH-style selection, Full PARC and the oracle diagnostic upper bound. All points are plotted using the same conservative label-uncertainty diagnostic. The figure emphasizes that threshold families can release aggressively but do not carry the PARC set-level certificate.

Extended Data Fig. 3: Ablation component heatmap. This heatmap shows release count, conservative FTR diagnostic and feasibility mass for component ablations. Rows include Full PARC, no verified-positive removal, post-filter without SCS, conservative empty-block policy, coverage-conditional policy, global calibration only, random audit-positive removal and wrongly removing uncertain cases. Actual frozen-matrix reruns and derived sensitivities are marked separately.

Extended Data Fig. 4: Null-inflation sensitivity. This display shows how release count and mass ratio change as verified-positive information is removed and uncertain-rate inflation is increased. It supports the claim that null-superset conservatism has a power cost: retaining more true-but-unverified paths in calibration raises block maxima and lowers release feasibility.

Extended Data Fig. 5: Non-exchangeability shifts. This display shows dataset transfer and covariate shifts using reader-facing labels. Actual reruns and derived sensitivities are indicated by different markers. The figure is interpreted as an assumption-boundary diagnostic rather than as a new benchmark comparison.

Extended Data Fig. 6: Audit-noise and score-miscalibration sensitivity. This display combines verified-positive perturbation and score-transformation sensitivity. False-to-true audit flips are highlighted as the validity-relevant direction because they can incorrectly remove false paths from the calibration null. True-to-false flips are power-relevant because they keep additional true paths in the null superset. Score transformations are labelled as monotone or noisy to distinguish rank-preserving and rank-disrupting changes.

Extended Data Fig. 7: Human-audit sample composition and agreement. This display contains Audit2000 label composition and the Boundary-500/Audit2000 reannotation agreement summaries. Verified-positive agreement is reported alongside the three-way label agreement. The caption explicitly states that the result is an audit-workflow reliability check, not a substitute for the formal risk theorem.

Extended Data Fig. 8: LVIS single-frame scoped evidence. This display shows the single-frame LVIS detection rows. GroundingDINO rows are plotted as scoped positive evidence under the official-unsupported proxy. OWLv2 rows are plotted as stress evidence because high-score unmatched audit labels remain pending. The caption does not describe LVIS as video tracking.

Extended Data Fig. 9: Mask-path proof-of-principle rows. This display shows box-to-mask or mask-path scaffold rows. It is captioned as proof-of-principle evidence, not a full official mask benchmark. Rows document how the release logic would be carried to a mask-path setting when the conversion and official benchmark provenance are complete.

Extended Data Fig. 10: Visual release/refusal examples. This display uses paper-facing visual assets to show the candidate units behind the statistical rows: adjacent-frame CTC cell links, SpaceNet same-building correspondences and camera-trap detections. The examples are illustrative visualizations of release and refusal decisions, not additional evidence for the guarantees. Public-domain wildlife images are used as schematic substitutes for iWildCam visual modes: species-level localized prompts can fail one-sided reliability, whereas animal-present detections are the unit used in the project-team-audited operational release.

Extended Data Fig. 11: Success-domain map. This display summarizes release/refusal outcomes against measurable certification features such as evidence mass, coverage, raw risk and conflict density. It is descriptive and is not used as a predictive theorem.

Extended Data Fig. 12: Release-certification benchmark-card map. This display maps reusable release-card tracks to the downstream artifacts they govern. It records the frozen universe, positive-support rule, block definition, risk level, decision and claim scope for each completed track.

Extended Data Fig. 13: Block-size heterogeneity diagnostic. This display visualizes size-stratified p-value diagnostics and block-size stress rows. Only the materials rows are completed candidate-level reruns; CTC and SpaceNet entries are labelled as aggregate or audit-sample diagnostics.

Extended Data Table 2: Reproducibility and provenance checklist. This table lists the exact review-safe artifacts included in the supplementary archive, the commands that rebuild each manuscript display, and the artifacts excluded because of benchmark licenses or model-cache restrictions. It also records SHA256 manifest coverage.

Extended Data Table 3: Release-certification benchmark tracks. This table summarizes the reusable release-card tracks. It is a governance index for the supplementary release cards rather than an additional experiment.

Extended Data Table 4: Prospective materials validation protocol status. This table records why A1 temporal splitting, A2 independent DFT cross-validation and the A3 prospective DFT follow-up pilot are protocol or feasibility rows. It makes explicit that A3 is frozen but blocked by a missing unlabeled generated candidate pool and has no selected candidates, no DFT job manifest and no DFT results.

Supplement K: Supplementary Data schema and provenance sidecars

The Supplementary Data files are machine-readable and independently checkable. Every row contains enough information to identify the dataset, proposal source, risk level, candidate budget, seed, release decision and provenance class. The schema is intentionally redundant: several columns repeat values used in the manuscript so that figures can be regenerated without consulting the main PDF.

Table 39: Common schema fields across release and stress CSVs.

Field family	Fields and meaning
Protocol identifiers	Dataset, task, generator, policy, risk level α , budget K , seed and candidate-universe identifier.
Release outputs	Release count, non-empty indicator, safe-refusal indicator, empty reason and decision label.
Risk diagnostics	UTR, audited FTR, conservative FTR, conservative unknown-as-false rate and audit coverage status.
Evidence diagnostics	Mass ratio, maximum observed e-value, required e-value threshold, selected e-value minimum/mean/maximum and self-consistency margin.
Provenance	Experiment basis, actual rerun versus derived sensitivity flag, source table id, source manifest hash and notes.
Scope	Main, appendix, stress, scoped generality, proof-of-principle or provenance-only placement.

Table 40: Provenance sidecar fields.

Field	Meaning
Source command	Command or script family that generated the row.
Input manifest	SHA256 manifest for input derived tables or candidate matrices.
Output manifest	SHA256 manifest for generated table, figure or diagnostic.
Environment	Python version, package lock file and plotting stack when relevant.
Raw-data requirement	Whether regenerating the row requires raw benchmark videos, raw annotations, model weights or detector caches.
Anonymization status	Whether path names, usernames, hostnames and repository identifiers have been stripped.
License restriction	Whether the artifact can be redistributed publicly or only described by provenance.

Release summary schema. The main release summary file records each dataset-generator row at the fixed main protocol. Required fields include dataset, generator, risk level, K , seeds, non-empty seeds, safe-refusal seeds, PARC mean release, raw top- K mean release, UTR, conservative label-uncertainty FTR, mass ratio, minimum mass ratio, safe-refusal rate and dominant refusal reason. Rows with empty release report conservative FTR as empty-release not applicable in the manuscript table, even though the mathematical empty-set FDP is zero.

Baseline summary schema. The baseline summary file aggregates method families. Required fields include baseline name, number of rows, dataset list, mean release and conservative FTR diagnostic. The source rows are grouped only after confirming that they use the same conservative diagnostic denominator. The file is used to plot the risk–utility frontier and to state the raw/threshold aggregate comparison in the main text.

Ablation schema. The ablation file records component name, baseline family, experiment basis, release count, risk diagnostics, mass ratio, maximum e-value, runtime where available and notes. Rows with experiment basis `actual_frozen_ablation_matrix` are actual reruns on frozen matrices. Rows with `derived_component_sensitivity` are sensitivity calculations from the Full PARC matrix. The manuscript distinguishes these categories.

Stress schema. Stress files contain stress-family-specific parameters. Null-inflation rows include label keep rate, verified-positive removal rate and uncertain-rate inflation. Non-exchangeability rows include shift scenario, split rule and assumption status. Audit-noise rows include noise type and noise rate. All stress files include release, conservative FTR, mass ratio, empty reason and result status.

Audit schema. Audit summary files contain dataset, label type and count. The updated audit closeout summaries contain row count, three-way label agreement, Cohen’s κ , disagreement count, exact verified-positive agreement and review status. If full row-level audit labels are distributed separately, the row-level schema contains anonymized row id, dataset, path id, review round, three-way label, verified-positive flag, uncertainty reason and SHA256 provenance for the visual evidence bundle.

Public-safe archive schema. The public-safe archive excludes real author names, e-mail addresses, local home directories, machine hostnames, non-anonymous repository names and unredacted absolute paths. The manuscript and data package use anonymized review identifiers.

Supplement L: Additional proof details and edge cases

This note expands the compact proof in Supplement A into the edge cases most likely to arise during review: ties in block ranks, conditioning on the fixed universe, pooled versus per-row interpretation, empty releases, and data-dependent selection. The statements below are not additional assumptions; they make explicit the conventions used by the main theorem.

L.1 Rank super-uniformity with ties

Let $Z_1, \dots, Z_n, Z_{n+1}$ be exchangeable calibration and test block scores, where Z_{n+1} is the block score associated with an actual false test path. Define the conservative rank p-value

$$q = \frac{1 + \sum_{i=1}^n \mathbf{1}[Z_i \geq Z_{n+1}]}{n + 1}.$$

For any $u \in [0, 1]$, the event $q \leq u$ implies that at most $\lfloor u(n + 1) \rfloor - 1$ of the n calibration blocks are at least as large as the test block. Equivalently, the test block must lie among the largest $\lfloor u(n + 1) \rfloor$ blocks under the conservative tie convention. Exchangeability makes the test index uniformly distributed over the $n + 1$ positions after random tie-breaking, and the non-random conservative tie rule can only increase q . Therefore $\Pr(q \leq u) \leq u$. This is the rank-super-uniform lemma used in the block p-value proof.

L.2 Conditioning on a frozen proposal universe

All validity statements are conditional on the event that the proposal universe has been fixed. Formally, let $\mathcal{G}$ be the sigma-field generated by the proposal backend, prompt set, score rule, linking rule, candidate budget, release grid, graph conflicts and tuning decisions. The proof is read conditionally on $\mathcal{G}$. After conditioning, the remaining randomness is the calibration/test sampling and latent validity labels under the stated exchangeability model. Conditioning on the observed verified-positive set is also part of this fixed-universe view: validity requires that the scores of actual false candidates within covered blocks remain exchangeable after the one-sided positives have been removed, because targeted verification changes the null-superset composition but may not remove false candidates. The theorem therefore certifies release for the realized finite universe $\mathcal{P}$, not for an unbounded space of objects that the proposal source did not generate.

L.3 False-only block and null-superset block

The proof compares two p-values. The unobservable false-only p-value uses

$$M_i^F = \max_{p \in \mathcal{F}_i} S_p,$$

where $\mathcal{F}_i = \{p : Y_p = 0\}$. The observable null-superset p-value uses

$$B_i^{\text{ns}} = \max_{p \in \mathcal{N}_i} S_p,$$

where $\mathcal{N}_i = \{p : A_p = 0\}$. Because $A_p = 1 \Rightarrow Y_p = 1$, every false path has $A_p = 0$, so $\mathcal{F}_i \subseteq \mathcal{N}_i$. Thus $B_i^{\text{ns}} \geq M_i^F$, and the null-superset p-value is at least the false-only p-value. The construction may lose power when true but unverified paths have high scores, but it does not become anti-conservative through this inclusion.

L.4 Empty blocks

An empty null-superset block can occur when no eligible unverified candidate exists in a cell/checkpoint. Two conventions are used. The conservative convention sets $B_i^{\text{ns}} = +\infty$, causing the block to count as exceeding every finite test score. This is marginally valid but can be powerless. The coverage-conditional convention removes the uncovered cell/checkpoint from the covered rank set and states the theorem inside the covered regime. Its additional assumption is conditional exchangeability after conditioning on coverage. The main text reports this distinction because the coverage convention supplies useful release power while the $+\infty$ convention supplies the conservative fallback.

L.5 E-calibrator details

The power calibrator $f_\gamma(u) = \gamma u^{\gamma-1}$, $0 < \gamma < 1$, satisfies

$$\int_0^1 f_\gamma(u) du = 1.$$

If U is super-uniform, then for any non-increasing nonnegative f with unit integral, $\mathbb{E}[f(U)] \leq 1$. The power calibrator is non-increasing for $0 < \gamma < 1$, so $E_p = f_\gamma(q_p^{\text{any}})$ is an e-value for every actual false path. The manuscript uses a tuning-fixed γ ; no test labels are used to choose it. Because the reported γ is a deterministic function of the frozen calibration resolution, replacing one fixed valid calibrator by this pre-evaluation deterministic choice preserves e-value validity under the same super-uniform p-value assumption.

L.6 Approximate e-value validity

The block-heterogeneity diagnostics also report a bounded-inflation interpretation. If a diagnostic or approximation argument establishes

$$\mathbb{E}[E_p] \leq 1 + \eta$$

for every actual false candidate rather than exact e-value validity, the same self-consistency proof yields an inflated expectation-level bound:

$$\mathbb{E}[\text{FTR}(R)] \leq \alpha(1 + \eta).$$

Indeed, the pointwise self-consistency inequality in Supplement L.7 gives

$$\frac{\mathbf{1}[p \in R]}{|R| \vee 1} \leq \frac{\alpha}{K} E_p$$

for each false candidate p . Summing over the at most K false candidates in the fixed universe and taking expectations gives $\alpha K^{-1} \sum_p \mathbb{E}[E_p] \leq \alpha(1 + \eta)$ when the same η bound applies to each false candidate. This lemma is not used to upgrade the main theorem; it records how bounded diagnostic inflation would translate into the same release selector.

L.7 Data-dependent release set

The SCS proof allows R to depend on all test e-values and graph conflicts. The only required final property is self-consistency. For each false path p ,

$$\mathbf{1}[p \in R] > 0 \quad \Rightarrow \quad E_p \geq \frac{K}{\alpha(|R| \vee 1)}.$$

Rearranging yields

$$\frac{\mathbf{1}[p \in R]}{|R| \vee 1} \leq \frac{\alpha}{K} E_p.$$

The inequality is pointwise in the realized data, so it remains valid even though R is selected after observing all e-values. Summing over false paths and taking expectations gives the FTR control statement.

L.8 Per-universe and pooled reporting

The theorem applies to the frozen evaluated run universe with candidate budget K . When the paper reports mean release across seeds or datasets, those are empirical summaries of repeated fixed-universe rows. The formal guarantee is attached to each row’s fixed universe under its own assumptions. If multiple independent universes are pooled into one release set, the proof can be applied to the pooled run universe with K equal to the total number of evaluated hypotheses and with e-values valid under the pooled calibration design. The manuscript does not rely on an unproved pooled guarantee across heterogeneous datasets; it reports row-level decisions and aggregates them descriptively.

L.9 Conservative diagnostics versus theorem target

The theorem target is the actual false-tracklet proportion

$$\frac{|R \cap \mathcal{H}_0|}{|R| \vee 1},$$

where $\mathcal{H}_0 = \{p : Y_p = 0\}$ is unobserved. The paper also reports UTR, audited FTR and conservative FTR diagnostics. These diagnostics are intentionally not treated as the theorem target. The conservative FTR diagnostic counts unsupported uncertain or unlabeled released paths as false in order to stress-test release behaviour under label uncertainty. It can therefore be larger than the actual false-tracklet proportion when many unsupported paths are visually true but unverified.

L.10 Finite-resolution feasibility

Let $E_{(1)} \geq E_{(2)} \geq \dots$ denote sorted e-values before graph conflicts. A uniform self-consistent set of size k can exist only if

$$E_{(k)} \geq \frac{K}{\alpha k}.$$

Equivalently,

$$\frac{\alpha k E_{(k)}}{K} \geq 1.$$

Thus $\max_k \alpha k E_{(k)} / M \geq 1$ is a necessary and sufficient condition for a non-empty uniform self-consistent frontier before compatibility constraints. This is a feasibility characterization, not an independent predictive model. Graph conflicts can only reduce the selected size relative to this frontier.

Proof edge cases and manuscript conventions. Ties use the conservative $\geq$ rank count; empty null blocks use the $+\infty$ fallback or covered-regime conditioning; data-dependent R only needs final self-consistency; graph conflicts are frozen before selection and can reduce power but not invalidate the proof; pooled summaries are descriptive unless a pooled universe is explicitly formed; conservative FTR is a label-uncertainty diagnostic, not the theorem target.